

Fixed Effects Regression Results

DERS Analysis

May 6, 2026

1 Overview

This note reports the results from the Stata fixed effects regression in `ders- analysis.do`. The analysis uses a simulated balanced panel with 100 units observed annually from 2015 through 2024, for a total of 1,000 observations. The data-generating process is

$$y_{it} = 1.5x_{it} + \alpha_i + \lambda_t + \varepsilon_{it}, \quad (1)$$

where α_i is a unit fixed effect, λ_t is a time effect, and ε_{it} is an idiosyncratic error term.

2 Specification

The estimated model is

$$y_{it} = \beta x_{it} + \lambda_t + \alpha_i + u_{it}. \quad (2)$$

The regression is estimated using Stata's `xtreg`, `fe` command with year fixed effects and standard errors clustered by unit:

```
xtreg y x i.year, fe vce(cluster id)
```

3 Results

Variable	Coefficient	Std. Error	<i>t</i> -statistic	<i>p</i> -value
<i>x</i>	1.5366	0.0329	46.74	0.000

Table 1: Fixed effects regression result for *x*

Statistic	Value
Observations	1,000
Groups	100
Within R^2	0.7257
Between R^2	0.2068
Overall R^2	0.5737
Clustered standard errors	Yes, by unit
Year fixed effects	Yes
Unit fixed effects	Yes

Table 2: Model summary

4 Interpretation

The estimated coefficient on x is 1.5366. Because the simulated true effect is 1.5, the fixed effects estimator recovers the underlying parameter closely. The coefficient is precisely estimated, with a clustered standard error of 0.0329 and a p -value below 0.001.

The within R^2 of 0.7257 indicates that the model explains a large share of within-unit variation in the dependent variable. The estimate is based on within-unit changes over time, while controlling for time-invariant unit characteristics and common year effects.

5 Conclusion

The Stata fixed effects regression confirms that the panel estimator recovers the simulated relationship between x and y . After accounting for unit fixed effects, year fixed effects, and clustered standard errors, the estimated effect of x remains close to the true value used to generate the data.